Jashanpreet Singh

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Summary

Computer Engineering undergraduate with hands-on experience in full-stack software development, backend API development and AI-powered applications. Strong foundation in DSA, OOPs, DBMS, and OS. Experienced in developing RESTful APIs, building data-processing workflows, creating test suites, and evaluating AI coding agents for software correctness and implementation quality. Seeking a Software Development Engineer role to contribute to reliable software solutions, efficient problem-solving, and continuous engineering improvement.

Work Experience

Software Intern, Defence Geoinformatics Research Establishment (DGRE), DRDO June 2026 – July 2026

- Developed **web-based software applications** for defence and geospatial research, contributing across **frontend development, API integration, and data visualization** workflows.
- Built reusable UI components and integrated **backend APIs** to support **2 key workflows**: geospatial data visualization and analytical processing.
- Collaborated with researchers and software teams across **multiple development activities** to deliver scalable, maintainable, and production-ready software solutions.

Projects

ScholarLogic | Smart AI-Powered Scholarship Platform Full-Stack + AI Integration

- Built a full-stack scholarship discovery platform that matches students with relevant opportunities based on eligibility criteria and student profiles.
- Developed a **Gemini 2.5 Flash ingestion pipeline** to extract scholarship eligibility rules and metadata from unstructured PDF documents.
- Integrated **Clerk, Vercel AI SDK, and Supabase** to implement authentication, contextual AI conversations, and persistent data management.
- Implemented scholarship document management and eligibility evaluation workflows to identify matching opportunities and explain why students may not qualify.

LiarLiar | AI-Driven Privacy Auditing & Network Interception React, TypeScript, Gemini API

- Developed a **Chrome Extension using Manifest V3** to compare website privacy-policy claims with observed network activity.
- Integrated the **Gemini API** to analyze natural-language privacy policies and convert data-collection claims into structured rules.
- Implemented a **deterministic rule-evaluation engine** to identify potential discrepancies between declared privacy practices and network requests.
- Configured Chrome's **declarativeNetRequest API** to block network requests that violate configured privacy-enforcement rules.

LocalLead Finder | Location-Based Business Lead Discovery Platform Next.js, TypeScript, API

- Built a full-stack business discovery platform supporting search by **business category, geographic region, location radius, and search criteria**.
- Integrated **Google Maps Places API** for location autocomplete, coordinate extraction, and location-based business discovery.
- Developed a **hexagonal grid-based search engine** to cover selected geographic areas with overlapping search cells and radius-based pruning.

Technical Skills

- **Programming Languages:** C++, C, Python, JavaScript, TypeScript
- **Backend & API Development:** Node.js, Express.js, FastAPI, RESTful API Development, API Integration, JWT Authentication
- **Frameworks & Libraries:** React, Next.js, OpenCV, Tailwind CSS
- **Databases:** SQL, MongoDB, PL/SQL, NoSQL
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems
- **Software Engineering:** Git, GitHub, Debugging, Data Analysis, Problem Solving, Technical Writing, Deployement

Education

Thapar Institute of Engineering and Technology B.E. in Computer Engineering 2023 – 2027

D.A.V. Public School, Lehragaga Senior Secondary Education (CBSE) 2023